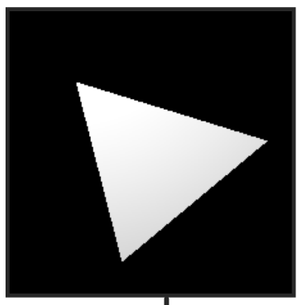
NeST decode tree on one real image: the model's greedy MAP path through the Hopf–HEALPix tree

nodes are true S^2 globes with real HEALPix cells; every branch probability is the trained model's actual 8-way softmax



actual input
(shape: tetrahedron)

root $p_0(c_0 \mid x)$

72-way · MAP $p = 0.10$

tokens condition every node

ViT-B/16 encoder

$\ell = 1$

8-way
$p(c_1 \mid \cdot)$
nside 2

$p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.07$ $p = 0.00$ $p = 0.93$

$\ell = 2$

8-way
$p(c_2 \mid \cdot)$
nside 4

$p = 0.99$ $p = 0.01$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$

$\ell = 3$

8-way
$p(c_3 \mid \cdot)$
nside 8

$p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.99$ $p = 0.00$ $p = 0.00$ $p = 0.00$

gold = the model's argmax child at each node (the MAP root→leaf path); branch width ∝ predicted probability. The chosen cell refines ×8 (pixel ×4, roll ×2) every level.